Name: Vallari Ravikiran Panvalkar
Roll No: 51
Experiment No.4
Develop a Responsive Photo Gallery Using CSS Grid, Hover Effects, and JavaScript
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Vidyavardhini's College of Engineering & Technology

Department of Computer Engineering

Aim:- To develop a responsive photo gallery using CSS Grid, hover effects, and JavaScript.

Objective: To design a responsive gallery layout using CSS Grid, implement hover effects for enhanced user interaction, and use JavaScript to add dynamic features such as image preview, filtering, or navigation.

Theory:

A responsive photo gallery is an important component of modern web design that allows images to be displayed in an organized and visually appealing manner across different devices. CSS Grid is a powerful layout system that enables developers to create flexible and structured gallery layouts with rows and columns. It automatically adjusts the placement and size of images based on screen size, ensuring responsiveness and consistency.

Hover effects using CSS enhance user interaction by providing visual feedback when users interact with images, such as zooming, fading, or highlighting. JavaScript is used to add dynamic functionalities like image preview in a modal, filtering images based on categories, and navigation through images. By combining CSS Grid, hover effects, and JavaScript, developers can create interactive and responsive photo galleries that improve user experience.

Implementation:

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Experiment 4 - Responsive Photo Gallery</title>

    <link rel="stylesheet" href="style.css">

</head>

<body>
```

```
<main class="page-shell">
    <section class="hero">
        <h1>Responsive Photo Gallery</h1>
        <p class="subtitle">
            A grid-based gallery with hover effects, category filters, and a JavaScript preview viewer.
        </p>
        <div class="filters" id="filters">
            <button class="filter-btn active" data-filter="all">All</button>
            <button class="filter-btn" data-filter="nature">Nature</button>
            <button class="filter-btn" data-filter="architecture">Architecture</button>
            <button class="filter-btn" data-filter="travel">Travel</button>
        </div>
    </section>


    <section class="gallery-section">
        <div class="gallery" id="gallery">
            <article class="photo-card" data-category="nature">
                <img src="https://images.unsplash.com/photo-1500530855697-b586d89ba3ee?auto=format&fit=crop&w=900&q=80" alt="Mountain landscape">
                <div class="overlay">
                    <h3>Mountain View</h3>
                    <p>Nature</p>
                </div>
            </article>


            <article class="photo-card" data-category="architecture">
```

```
            <img src="https://images.unsplash.com/photo-1511818966892-
d7d671e672a2?auto=format&fit=crop&w=900&q=80" alt="Modern architecture">

            <div class="overlay">

                <h3>Modern Structure</h3>

                <p>Architecture</p>

            </div>

        </article>



        <article class="photo-card" data-category="travel">

            <img src="https://images.unsplash.com/photo-1507525428034-
b723cf961d3e?auto=format&fit=crop&w=900&q=80" alt="Beach travel scene">

            <div class="overlay">

                <h3>Beach Escape</h3>

                <p>Travel</p>

            </div>

        </article>



        <article class="photo-card" data-category="nature">

            <img src="https://images.unsplash.com/photo-1511497584788-
876760111969?auto=format&fit=crop&w=900&q=80" alt="Forest trail">

            <div class="overlay">

                <h3>Forest Trail</h3>

                <p>Nature</p>

            </div>

        </article>



        <article class="photo-card" data-category="architecture">
```

```
            <img src="https://images.unsplash.com/photo-1486325212027-
8081e485255e?auto=format&fit=crop&w=900&q=80" alt="Skyscraper skyline">

            <div class="overlay">

              <h3>Skyline</h3>

              <p>Architecture</p>

            </div>

          </article>



          <article class="photo-card" data-category="travel">

            <img src="https://images.unsplash.com/photo-1473625247510-
8ceb1760943f?auto=format&fit=crop&w=900&q=80" alt="Travel road trip">

            <div class="overlay">

              <h3>Open Road</h3>

              <p>Travel</p>

            </div>

          </article>

        </div>



        <p class="hint">Click any image to view it larger.</p>

      </section>

    </main>



    <div class="lightbox" id="lightbox" aria-hidden="true">

      <button class="lightbox-close" id="closeLightbox" aria-label="Close
preview">&times;</button>

      <img id="lightboxImage" alt="Preview image">

      <div class="lightbox-caption" id="lightboxCaption"></div>
```

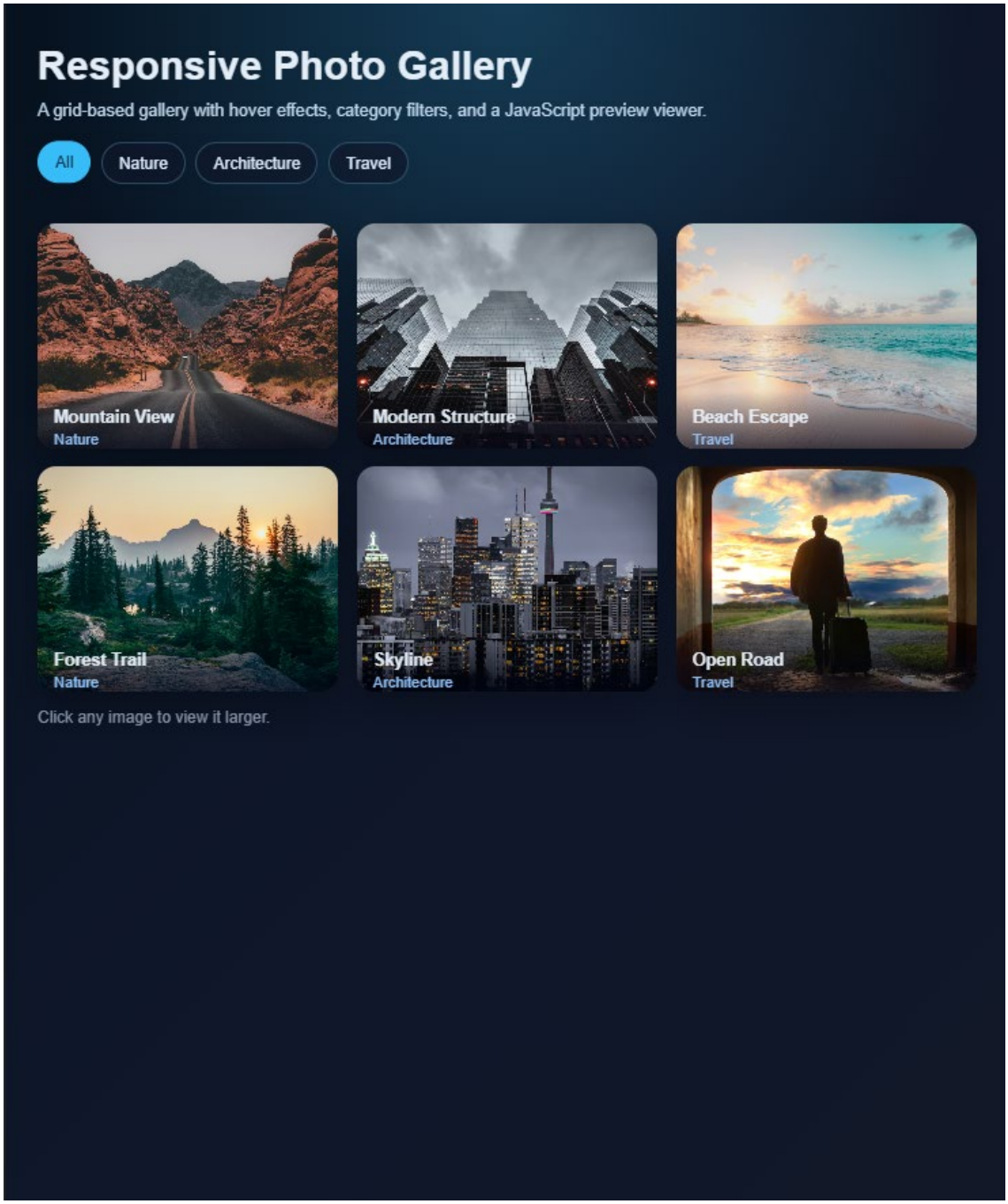
```
    </div>


    <script src="script.js"></script>

</body>

</html>
```

Output:



Conclusion:

Designing a responsive gallery using **CSS Grid** ensures a flexible and well-structured layout that adapts to different screen sizes. **Hover effects** enhance user interaction by making the gallery more engaging and visually appealing. With the help of **JavaScript**, dynamic features like image preview, filtering, and navigation improve functionality and user experience. Overall, this approach results in a modern, interactive, and mobile-friendly photo gallery